

Root Cause Analysis

Scenario-Focused Deep Dive — the RCA toolkit for Problem Management



Focus indicators



Multi-answer scenarios



Tool-relationship diagrams

Companion to your ITIL 4 Foundation, Incident Management & Process Improvement guides.

Go See · Fishbone · Is/Is Not · Pareto · Voting · 5 Whys · Therefore · Validate

How to Use This Guide

RCA is the engine of **Problem Management**. Scenario questions here rarely ask "define Pareto" — they describe a situation and ask **which tool fits**, **what order** to use them, or **which principles** are being followed or broken. Several answers can be correct. Learn what each tool **does**, and how they **combine**.

The Focus System

**MUST
MASTER**



Incident vs Problem link · 5 Whys · Fishbone · the tool-choice logic · "fix the system, not the person."

HIGH VALUE



Pareto (80/20) · Is/Is Not · Go See · Voting · Therefore validation.

SUPPORTING



Branching, best-practice tips, common root-cause patterns.

The one link the exam always tests 🔥🔥🔥

Incident Management

Restore normal service **as fast as possible**. A temporary fix or **workaround** is fine. Symptom-focused, speed-focused.

vs

Problem Management

Find & remove the **root cause** using **RCA** so it **never recurs**. Cause-focused, permanence-focused.

One-line memory: Incident = restore service fast. Problem = find the root cause so it never happens again.
RCA lives inside Problem Management.

RCA in one sentence: a structured, **fact-based** method to find the **underlying cause** of a problem so it can be **permanently resolved and prevented from recurring** — fix the **cause**, not the symptom.

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The tool map — which tool does what (memorise this table):

Tool	Job	One word
Go See	Observe the real process, gather facts	Observe
Fishbone	List all possible causes (brainstorm)	List
Is / Is Not	Narrow down — eliminate unlikely causes	Narrow
Pareto	Prioritise using data (80/20)	Prioritise
Voting	Team picks most likely causes to pursue	Select
5 Whys	Find the deepest, fixable cause	Deepen
Therefore	Verify the cause→effect logic	Verify
Validate	Confirm the problem doesn't recur	Confirm

1 RCA Fundamentals & Team Principles MUST MASTER 🔥🔥🔥

Symptom vs Root Cause

Symptom (what you see)	Root cause (why it happened)
Website crashes	Memory leak in application
Late deliveries	Incorrect inventory records
Slow computer	Faulty hard drive

Classic trap: patching symptoms (counting inventory monthly, blaming staff) leaves the problem alive. RCA fixes the **controllable, fundamental** cause so it stops recurring.

Working with others — the TEAM principles 🔥🔥

No one person sees the whole problem: Network says "network," DB says "database," App says "app." Together they find the truth.

T — Together

Involve multiple teams → complete picture + ownership.

E — Evidence

Facts, logs, metrics — not opinions ("I think it's the server").

A — Act quickly

Investigate soon — evidence is fresh.

M — Make the system better

Don't blame people — ask "why was the mistake possible?"

Fix the system, not the person. When a human "erred," look for the enabling cause: no training, poor process, missing documentation, bad system design. Blame stops learning.



Scenario (multi-answer)

After an outage, a manager wants to "restore service now and blame whoever ran the bad command." An engineer suggests a proper RCA later. Which statements align with ITIL/RCA principles? (select all that apply)

A. Restoring service fast is correct Incident Management (a workaround is fine).

B. Finding the root cause afterwards is Problem Management via RCA.

C. Blaming the individual is wrong — ask why the mistake was possible.

D. A fast restore means no root-cause work is needed.

Correct: A, B, C. D is wrong — restoring service doesn't remove the cause; that's what Problem Management/RCA is for.

2 Go See the Business Process HIGH VALUE 🔥🔥

Go to **where the problem occurs** and observe the actual work — don't guess from your desk.

Flow: Go → See → Ask → Map → Fix. Go to the workplace, See the process, Ask the people, Map the workflow, Fix the root cause.

Why

Avoid assumptions · find the real cause · understand how work actually flows · talk to doers.

Look for

Delays, errors, bottlenecks, handoff waits — then run RCA.

Worked example. "Loan approvals too slow." On-site observation finds documents wait 5 days for missing info and applications wait 3 days between departments. **Root causes:** poor inter-department communication + no measurement of waiting time. **Fix:** improve communication, track handoff times.

Ties to Process Improvement: "Go See" is how you find **bottlenecks and queues** in real life — the same delays that dominate lead time. Observation gives facts for the other RCA tools.

🧩 Scenario (multi-answer)

A team debates the cause of slow approvals in a meeting room, each blaming another department. What does "Go See" recommend? (select all that apply)

- A. Visit where the work happens and observe it step by step.
- B. Talk to the people actually doing the work.
- C. Map the process and find delays/handoff waits.
- D. Decide the cause from the meeting room using assumptions.

Correct: A, B, C. D is the exact anti-pattern "Go See" exists to prevent.

3 Fishbone Diagram MUST MASTER 🔥🔥🔥

Also called **Cause-and-Effect** or **Ishikawa** diagram. A visual brainstorm to identify and organise **all possible causes** — so none are overlooked.



Head = the problem · Bones = cause categories · Small bones = specific sub-causes.

Two things the exam checks: (1) Fishbone **does not identify** the root cause by itself — it **organises possibilities** for further investigation. (2) You **don't have to** use the classic "Man/Machine/Method/Material" categories — pick categories that fit **your** process (Communication, Process, Equipment, Training, Documentation...).

Tool relationships: Fishbone = find **all** possible causes · 5 Whys = find the **deepest** cause · Pareto = decide **what to fix first**. They work together, not instead of each other.

🧩 Scenario (multi-answer)

A hospital team builds a Fishbone for "patients misdiagnosed," listing many causes across Communication, Records, Lab Equipment and Doctors. Which statements are correct? (select all that apply)

- ☐ A. Fishbone organises all possible causes but doesn't prove the root cause.
- ☐ B. Categories should fit this process, not forced "4 Ms."
- ☐ C. Next they should narrow/prioritise (Voting, Is/Is Not, Pareto) then 5 Whys.
- ☐ D. The Fishbone alone tells them exactly which cause to fix.

Correct: A, B, C. D is wrong — Fishbone lists; it doesn't conclude.

4 Is / Is Not Analysis HIGH VALUE 🔥🔥

A questioning technique that **narrows** the list by comparing where the problem **is** vs **is not**. The **difference** between them often points at the cause.

Question	IS	IS NOT
Who?	Who is affected?	Who is not?
What?	What is affected?	What is not?
Where?	Where does it happen?	Where doesn't it?
When?	When did it start?	When did it not?
How much?	How many affected?	How many not?

Worked example. Some lotion boxes arrive damaged. **IS:** customers C/D/E, skin-rash lotion, after Oct 15, 30% of shipments. **IS NOT:** customers A/B, other products, before Oct 15, other 70%. The difference → after Oct 15 the firm switched to **weaker boxes** for heavier products. **Fix:** stronger boxes.

Memory: IS = where the problem exists · IS NOT = where it doesn't · the **difference** reveals the cause. Fact-based, eliminates unlikely causes, focuses RCA.

Scenario (multi-answer)

A defect appears only on the **second shift**, only on **one product line**, only **after** a date. Which are true about using Is/Is Not here? (select all that apply)

- ☐ A. It narrows causes by contrasting affected vs unaffected conditions.
- ☐ B. The "second shift only" difference is a strong lead to investigate.
- ☐ C. It's fact-based and eliminates unlikely causes.
- ☐ D. It proves the final root cause on its own.

Correct: A, B, C. D is wrong — it *narrows*; you still confirm with 5 Whys/observation.

5 Pareto Analysis (80/20) HIGH VALUE 🔥🔥

Identifies the **few causes responsible for most problems**, so you know what to fix first.

80% of problems come from 20% of causes

20% = "Vital Few" (fix first) · 80% = "Trivial Many" · numbers are a guideline, not exact

Worked example. Hotel complaints: Front Desk 620, Room Cleaning 270, Breakfast 230, Wi-Fi 180, A/C 25. The top three (Front Desk, Cleaning, Breakfast) $\approx$ 80% of complaints → fix those first for the biggest gain.

Steps

1. List problems → count frequency → sort high→low → cumulative % → focus on the ~80%.

Pareto = Prioritise. Data-driven, used in Problem Management & RCA to decide the order of attack.

🧩 Scenario (multi-answer)

A support team has 12 recurring incident categories and limited capacity. Data shows 3 categories cause ~78% of tickets. What does Pareto guidance say? (select all that apply)

A. Fix the "vital few" 3 categories first for the biggest impact.

B. Use data, not opinion, to prioritise.

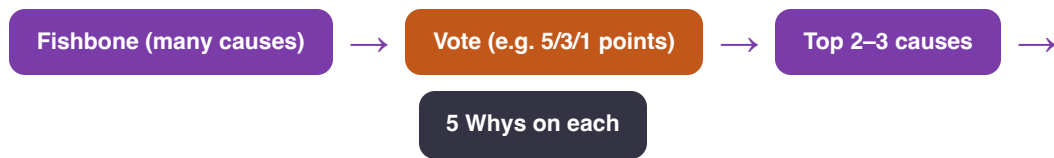
C. 80/20 is a guideline — 78/22 is fine.

D. Spread effort equally across all 12 categories.

Correct: A, B, C. D is wrong — equal effort ignores the whole point of Pareto.

6 Voting on Causes SUPPORTING 🔥

After a Fishbone lists **too many** causes, the team **votes** to pick the most likely 2–3 to investigate first.



Key exam point: Voting **does not prove** the root cause — it only chooses **where to investigate next**. Base votes on facts; include people who know the process; keep the full Fishbone for later.

🧩 Scenario (multi-answer)

A Fishbone for "misdiagnosis" has 15 causes. The team votes; "overburdened doctors" and "lack of test equipment" score highest. Which statements are correct? (select all that apply)

- A. Voting prioritises which causes to investigate next.
- B. The next step is 5 Whys on the top-voted causes.
- C. Voting itself confirms the root cause.
- D. Votes should be based on facts and process knowledge.

Correct: A, B, D. C is wrong — voting selects, it doesn't confirm.

7 5 Whys + Therefore + Branching MUST MASTER 🔥🔥🔥

Repeatedly ask **"Why?"** until you reach a **fixable** root cause. **"5" is a guideline**, not a rule — stop when the cause is controllable and fixable.



Worked example. Irregular cash flow → most sales at quarter end → customers wait for discounts → discounts only at quarter end → company chases quarterly targets → **performance is measured quarterly** (root cause).

Validate with "Therefore" 🔥🔥

Read the chain **forward**: "Performance measured quarterly → **therefore** discounts at quarter end → **therefore** customers wait → **therefore** late sales → **therefore** irregular cash flow." If every "therefore" makes sense, the logic holds.

Branching — one problem, several causes SUPPORTING

Normal 5 Whys

One chain → one root cause → one solution.

vs

Branching 5 Whys

Multiple chains → multiple root causes → multiple solutions.

Branching example. "Repairs late" → wrong serial number. **Branch 1:** poor lighting → wrong bulb → manual missing bulb details (fix the manual). **Branch 2:** operator untrained → no unpacking procedure (create process + training). **Both fixes needed.**

🧩 Scenario (multi-answer)

Investigating a recurring defect, the team finds two independent causal chains and asks "why" only 3 times on one before reaching a fixable cause. Which statements are correct? (select all that apply)

- A. Stopping at 3 "whys" is fine — 5 is a guideline.
- B. Two chains means branching — expect multiple root causes/solutions.
- C. Validate each chain with "therefore" logic.
- D. Every problem must have exactly one root cause.

Correct: A, B, C. D is wrong — problems can have multiple root causes (that's why branching exists).

8 Combining Tools & Best Practices MUST MASTER 🔥🔥🔥

No single tool finds every root cause. The power is in the **sequence**.

The typical RCA flow



Three very common root causes (process-related) 🔥🔥

No standard process exists.

Standard exists but not followed.

Standard is unclear / incomplete (ambiguous).

Best-practice mindset (from the Q&A)

- **Biggest mistake:** starting without a clearly defined problem ("a well-defined problem is half solved").
- **Don't blame people** — fix process/policy/procedure/training/communication.
- **"No time for RCA"** is wrong — you don't have time *not* to; it stops repeat firefighting.
- **Communication** is a frequent root cause (unclear instructions, assumptions, jargon, missing info).
- **Validate** the fix: monitor; if it recurs, the true cause wasn't found.



Scenario (multi-answer)

A team applies a fix, but the incident returns after 6 weeks. They also admit they never wrote a clear problem statement. What went wrong / what's true? (select all that apply)

A. Recurrence means the true root cause wasn't fully identified.

B. Validation (monitoring) correctly caught the failure.

C. Skipping a clear problem statement undermined the whole RCA.

D. Recurrence proves RCA is a waste of time.

Correct: A, B, C. D is wrong — recurrence shows RCA must continue, not that it's pointless.

9 Scenario Question Bank TRAIN HERE 🔥🔥🔥

Q1 · Pick the right tool

A team has just brainstormed 20 possible causes on a whiteboard and needs to decide which 2–3 to investigate with limited time and little hard data. Which tools fit **now**? (select all that apply)

- A. Voting (team picks most likely)
- B. Is/Is Not (narrow by contrast)
- C. Pareto (if they had frequency data)
- D. Start building the product fix immediately

Correct: A, B, C. With data, Pareto; without, Voting/Is-Is-Not. D skips finding the cause.

Q2 · Incident vs Problem

Email is down. The team restores it in 10 min with a restart, then opens an investigation into the recurring failure. Which are correct? (select all that apply)

- A. The restart is Incident Management (restore fast, workaround OK).
- B. The investigation is Problem Management using RCA.
- C. RCA aims to prevent recurrence, not just restore.
- D. The restart already performed root cause analysis.

Correct: A, B, C. D is wrong — a restart restores service, it doesn't find the cause.

Q3 · Tool relationships

Which statements about how RCA tools relate are correct? (select all that apply)

- A. Fishbone lists all possible causes.
- B. 5 Whys finds the deepest fixable cause of a selected branch.
- C. Pareto prioritises causes using data.
- D. Fishbone by itself identifies the single root cause.

Correct: A, B, C. D is wrong — Fishbone organises; it doesn't conclude.

Q4 · Culture & principles

An engineer "typed the wrong command." Which responses match RCA best practice? (select all that apply)

- A. Ask why the wrong command was possible (missing checks, no training?).
- B. Improve the system/process to prevent it (Poka-Yoke-style guardrails).
- C. Use facts/logs, not blame.
- D. Discipline the engineer and close the case.

Correct: A, B, C. D is wrong — blame fixes nothing and stops learning.

Q5 · 5 Whys mechanics

Which statements about 5 Whys are correct? (select all that apply)

- A. "5" is a guideline; stop at the fixable cause.
- B. Validate the chain forward with "therefore."
- C. A problem may branch into multiple root causes.
- D. You must always ask exactly five whys.

Correct: A, B, C. D is wrong — the number is flexible.

Q6 · Validation

After a countermeasure, which actions/beliefs are correct? (select all that apply)

A. Monitor the process (e.g. 30–90 days) for recurrence.

B. No recurrence suggests the root cause was fixed.

C. Recurrence means the true cause wasn't fully identified.

D. Once a fix is deployed, no follow-up is needed.

Correct: A, B, C. D is wrong — validation is essential to confirm the fix.

10 Rapid Revision & Tool Chooser

Incident vs Problem 🔥🔥🔥

Incident = restore fast (workaround OK). Problem = remove root cause via RCA so it never recurs.
RCA lives in Problem Management.

RCA definition 🔥🔥🔥

Structured, fact-based method to find the underlying cause and permanently prevent recurrence. Fix the cause, not the symptom.

Tool jobs 🔥🔥🔥

Go See = observe · Fishbone = list all · Is/Is Not = narrow · Pareto = prioritise (80/20) · Voting = select · 5 Whys = deepen · Therefore = verify · Validate = confirm.

Typical order 🔥🔥

See → Fishbone → Is/Is Not → Pareto → (Voting) → 5 Whys → Therefore → Fix → Validate.

Team = TEAM 🔥🔥

Together · Evidence · Act quickly · Make the system better (no blame).

5 Whys rules 🔥🔥🔥

"5" is a guideline · validate with "therefore" · branch when multiple causes · stop at a fixable cause.

3 common root causes 🔥🔥

No standard process · standard not followed · standard unclear/incomplete. Communication is a frequent cause.

Golden rules ✅

Define the problem clearly · facts not opinions · multiple teams · fix system not people · combine tools · verify logic · confirm no recurrence.

Tool chooser (scenario shortcut):

"Don't know the cause, haven't looked" → **Go See**. "Need every possibility" → **Fishbone**.

"Too many possibilities, no data" → **Voting / Is/Is Not**. "Have frequency data, what first?" → **Pareto**.

"Have a candidate cause, go deeper" → **5 Whys**. "Check my logic" → **Therefore**. "Did it work?" → **Validate**.

Exam-day tactic for "select all that apply": the wrong option is usually an **overclaim** or a **blame/shortcut** — "Fishbone identifies the root cause," "voting confirms the cause," "must ask exactly 5 whys," "every problem has one cause," "discipline the person," "no follow-up needed." RCA always prefers **observe, narrow, deepen, verify, validate** — and fix the system.